VISION	MISSION
To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.	1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities. 2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program. 3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Practical No. 00

Aim: Introduction to Machine Learning Lab.

Software Required: Python IDE, Anaconda, VS Code, Jupyter Nootbook, Google Colab.

Theory:

Introduction to Machine Learning:

In the real world, we are surrounded by humans who can learn everything from their experiences with their learning capability, and we have computers or machines which work on our instructions. But can a machine also learn from experiences or past data like a human does? So here comes the role of Machine Learning.

A subset of artificial intelligence known as machine learning focuses primarily on the creation of algorithms that enable a computer to independently learn from data and previous experiences. Arthur Samuel first used the term "machine learning" in 1959. It could be summarized as follows:

Without being explicitly programmed, machine learning enables a machine to automatically learn from data, improve performance from experiences, and predict things.

Machine learning algorithms create a mathematical model that, without being explicitly programmed, aids in making predictions or decisions with the assistance of sample historical data, or training data. For the purpose of developing predictive models, machine learning brings together statistics and computer science. Algorithms that learn from historical data are either constructed or utilized in machine learning. The performance will rise in proportion to the quantity of information we provide.

The early history of Machine Learning (Pre-1940):

- **1834:** In 1834, Charles Babbage, the father of the computer, conceived a device that could be programmed with punch cards. However, the machine was never built, but all modern computers rely on its logical structure.
- **1936:** In 1936, Alan Turing gave a theory that how a machine can determine and execute a set of instructions.

The era of stored program computers:

- **1940:** In 1940, the first manually operated computer, "ENIAC" was invented, which was the first electronic general-purpose computer. After that stored program computer such as EDSAC in 1949 and EDVAC in 1951 were invented.
- **1943:** In 1943, a human neural network was modeled with an electrical circuit. In 1950, the scientists started applying their idea to work and analyzed how human neurons might work.

Computer machinery and intelligence:

- **1950:** In 1950, Alan Turing published a seminal paper, "**Computer Machinery and Intelligence**," on the topic of artificial intelligence. **In his paper, he asked, "Can machines think?"**

Machine intelligence in Games:

- **1952:** Arthur Samuel, who was the pioneer of machine learning, created a program that helped an IBM computer to play a checker's game. It performed better more it played.
- **1959:** In 1959, the term "Machine Learning" was first coined by **Arthur Samuel**.

The first "AI" winter:

- The duration of 1974 to 1980 was the tough time for AI and ML researchers, and this duration was called as **AI winter**.
- In this duration, failure of machine translation occurred, and people had reduced their interest from AI, which led to reduced funding by the government to the researches.

Need for Machine Learning

The demand for machine learning is steadily rising. Because it is able to perform tasks that are too complex for a person to implement directly, machine learning is required. Humans are constrained by our inability to manually access vast amounts of data; as a result, we require computer systems, which is where machine learning comes in to simplify our lives.

By providing them with a large amount of data and allowing them to automatically explore the data, build models, and predict the required output, we can train machine learning algorithms. The cost function can be used to determine the amount of data and the machine learning algorithm's performance. We can save both time and money by using machine learning.

The significance of AI can be handily perceived by its utilization's cases, Presently, AI is utilized in self-driving vehicles, digital misrepresentation identification, face acknowledgment, and companion idea by Facebook, and so on. Different top organizations, for example, Netflix and Amazon have constructed AI models that are utilizing an immense measure of information to examine the client interest and suggest item likewise.

Following are some key points which show the importance of Machine Learning:

- Rapid increment in the production of data
- Solving complex problems, which are difficult for a human
- Decision making in various sector including finance
- Finding hidden patterns and extracting useful information from data.

Classification of Machine Learning

At a broad level, machine learning can be classified into three types:

1. **Supervised learning**
2. **Unsupervised learning**
3. **Semi- Supervised learning**
4. **Reinforcement learning**

1) Supervised Learning

In supervised learning, sample labeled data are provided to the machine learning system for training, and the system then predicts the output based on the training data.

The system uses labeled data to build a model that understands the datasets and learns about each one. After the training and processing are done, we test the model with sample data to see if it can accurately predict the output.

The mapping of the input data to the output data is the objective of supervised learning. The managed learning depends on oversight, and it is equivalent to when an understudy learns things in the management of the educator. Spam filtering is an example of supervised learning.

Supervised learning can be grouped further in two categories of algorithms:

- **Classification**
- **Regression**

2) Unsupervised Learning

Unsupervised learning is a learning method in which a machine learns without any supervision.

The training is provided to the machine with the set of data that has not been labeled, classified, or categorized, and the algorithm needs to act on that data without any supervision. The goal of unsupervised learning is to restructure the input data into new features or a group of objects with similar patterns.

In unsupervised learning, we don't have a predetermined result. The machine tries to find useful insights from the huge amount of data. It can be further classifieds into two categories of algorithms:

- **Clustering**
- **Association**

3) Semi-supervised learning:

Where an incomplete training signal is given: a training set with some (often many) of the target outputs missing. There is a special case of this principle known as Transduction where the entire set of problem instances is known at learning time, except that part of the targets are missing. Semi-supervised learning is an approach to machine learning that combines small labeled data with a large amount of unlabeled data during training. Semi-supervised learning falls between unsupervised learning and supervised learning.

4) Reinforcement Learning

Reinforcement learning is a feedback-based learning method, in which a learning agent gets a reward for each right action and gets a penalty for each wrong action. The agent learns automatically with these feedbacks and improves its performance. In reinforcement learning, the agent interacts with the environment and explores it. The goal of an agent is to get the most reward points, and hence, it improves its performance.

The robotic dog, which automatically learns the movement of his arms, is an example of Reinforcement learning.

Applications of Machine Learning

Machine Learning is used in many industries to solve problems and improve services. Here are some common real-world applications:

1. **Healthcare:** It helps doctors to diagnose diseases from medical images like X-rays and MRIs. It also predicts patient outcomes and personalizes treatments which improves healthcare quality.

2. **Finance:** In finance it detects fraudulent transactions in real time and supports algorithmic trading. It also helps to assess credit risk helps in making lending safer and faster.

3. **Retail and E-Commerce:** It helps in personalized product recommendations and forecasts demand to optimize inventory and also analyzes customer sentiment to improve shopping experiences.

4. **Transportation and Automotive:** Self-driving cars rely on ML to navigate and make decisions. It optimizes delivery routes and predicts vehicle maintenance needs which reduces downtime.

5. **Social Media and Entertainment:** Platforms like Netflix and YouTube use ML to recommend content we'll enjoy. It enables image and speech recognition for better user interaction.

6. **Manufacturing:** It improves quality control by detecting defects in products automatically and predicts machine failures in advance and helps in production processes.

Advantages of Machine Learning:

1. **Automation of Tasks**
   ML can automate repetitive and time-consuming tasks, reducing the need for human intervention.
2. **Continuous Improvement**
   ML models improve their performance over time as they are exposed to more data.
3. **Efficient Data Handling**
   It can process and analyze large volumes of data faster and more accurately than humans.
4. **Wide Range of Applications**
   Used in diverse fields: healthcare (disease prediction), finance (fraud detection), marketing (customer segmentation), etc.
5. **Pattern Recognition**
   ML can detect hidden patterns or trends in complex datasets that are not obvious to humans.

6. **Real-Time Decision Making**
   Enables fast and accurate decisions in real-time systems like recommendation engines and self-driving cars.

Disadvantages of Machine Learning:

1. **Data Dependency**
   Requires large, high-quality datasets to perform well. Poor data leads to poor outcomes.
2. **High Computational Cost**
   Training complex models, especially deep learning ones, can be resource-intensive and expensive.
3. **Lack of Transparency (Black Box Models)**
   Some models (like neural networks) are hard to interpret and understand.
4. **Overfitting & Underfitting**
   Models can become too specific or too general, reducing their real-world accuracy.
5. **Bias in Data**
   If training data is biased, the model will also be biased, leading to unfair or inaccurate results.
6. **Security Concerns**
   Vulnerable to adversarial attacks or data manipulation, especially in critical systems.

Introduction about Google Colab:

Google Colab (short for Colaboratory) is a free, cloud-based platform provided by Google that allows users to write and execute Python code through the browser. It is especially popular for machine learning, data analysis, and educational purposes because it supports powerful computing resources such as GPUs and TPUs.

It is built on Jupyter Notebook, so it provides an interactive coding environment where users can write code, add text, and visualize results all in one place.

Key Features:

- Free access to GPUs and TPUs for faster computation.
- No installation needed – runs entirely in the browser.
- Seamless integration with Google Drive for saving and sharing notebooks.
- Supports libraries like TensorFlow, PyTorch, NumPy, pandas, etc.
- Ideal for collaborative projects, similar to how Google Docs works.

Conclusion: This practical provided a foundational understanding of machine learning concepts, including supervised and unsupervised learning. It enhanced my technical skills and prepared me for implementing more advanced algorithms in future experiments.